

Summary

My research interests lie in **Medical Image Analysis, Computer Vision, and Generative AI**. I focus on developing time-efficient and robust machine learning algorithms for image registration, motion tracking, medical imaging synthesis, and atlas construction, with applications in neuroimaging and cardiovascular imaging. My research has been published in leading venues, including MICCAI, IPMI, ISBI, and the Machine Learning for Biomedical Imaging (MELBA) journal.

Education

Ph.D. in Computer Engineering	University of Virginia Virginia, USA	08/2023-05/2028
M.S. in Computer Technology	East China Normal University Shanghai, China	09/2020-07/2023

Publications

- [1] **Wu, N.**, Jayakumar N, Xing J., Zhang M.. *Geodesic-informed Generative Diffusion Model for Topology-preserved Image Video Generation. The Journal of Machine Learning for Biomedical Imaging, MELBA (Under Review).*
- [2] **Wu, N.** and Zhang M.. *DeepLDDMM: Learning-Based Large Deformation Metric Mappings with Geodesic Constraint. IEEE International Symposium on Imaging, ISBI2026 (Under Review).*
- [3] Deb, S., **Wu N.**, Epstein F.H., Zhang M.. *Unsupervised Cardiac Video Translation Via Motion Feature Guided Diffusion Model. 28th International Conference on Image Computing and Computer Assisted Intervention, MICCAI2025 (Oral).*
- [4] **Wu, N.** and Zhang, M., *Learning Geodesics of Geometric Shape Deformations from Images. The Journal of Machine Learning for Biomedical Imaging, MELBA (Accepted, 2026).*
- [5] **Wu, N.**, Jayakumar N, Xing J., Zhang M.. *IGG: Image Generation Informed by Geodesic Dynamics in Deformation Spaces. International Conference on Information Processing and Image Analysis, IPMI2025 (Oral).*
- [6] **Wu, N.**, Xing J., and Zhang M.. *TLRN: Temporal Latent Residual Networks for Large Deformation Image Registration. 27th International Conference on Image Computing and Computer Assisted Intervention, MICCAI2024 (Accepted).*
- [7] Xing J.R., Jayakumar N, **Wu N.**, Bilchickd K., Epstein F. and Zhang M. *LaMoD: Latent Motion Diffusion Model for Myocardial Strain Generation. International Workshop on Shape in Medical Imaging, ShapeMI 2024 (Accepted).*
- [8] Xing J.R., **Wu N.**, Bilchickd K., Epstein F. and Zhang M. *Multimodal Learning to Improve Cardiac Late Mechanical Activation Detection from Cine MR Images. IEEE International Symposium on Imaging, ISBI2024 (Oral).*
- [9] **Wu, N.** and Zhang, M.. *NeurEPDiff: Neural Networks to Predict Geodesics in Deformable Spaces. International Conference on Information Processing and Image Analysis, IPMI2023 (Oral).*
- [10] **Wu, N.**, Wang, J., Zhang, M., Zhang, G. X., Peng, Y. X., and Shen, C. M. *Hybrid Atlas Building with Deep Registration Priors. IEEE International Symposium on Image Processing, ISBI2022 (Accepted).*

Research Experience

Department of Computer Engineering, University of Virginia

Charlottesville, VA, USA

Graduate Research Assistant, Advisor, Prof. Miaomiao Zhang

08/2023-present

- **Geodesic Modeling & Registration.** Developed a neural operator that for the first time fast predicts the geodesics in deformation spaces governed by a well-known Euler-Poincaré differential equation (EPDiff). The neural operator is specifically designed to learn the mapping function, EPDiff, which defines the evolving trajectory of geodesic deformations parameterized in the tangent space of diffeomorphisms (a.k.a velocity fields). This method has been applied to brain magnetic resonance imaging (MRI) registration. Related publications include [1], [2], [4], [5], and [9].
- **Temporal Image Registration.** Developed a temporal latent residual network (TLRN) for time-series image

registration. TLRN leverages the nature of motion continuity and exploits the temporal smoothness in consecutive image frames, achieving substantially improved registration accuracy and smoothness in cine cardiac magnetic resonance (CMR) videos registration. Related publications include [6] and [3].

• **Generative AI & Topology-Preserved Synthesis.** Introduced a geodesic-informed generative diffusion model for topology-preserved image video generation, which the first time integrates topology-preserving geodesic principles into the diffusion-based generative process. In contrast to conventional methods that operate in image intensity space, our model learns and synthesizes diverse samples within geodesic deformation spaces, where geometric object changes are learned as smooth and invertible smooth mappings from a given template/source image. This method has been applied to brain MRI generation and shown to improve downstream segmentation performance. Related publications include [1] and [5].

Department of Computer Technology, East China Normal University

Shanghai, China

Graduate Research Assistant, Advisor, Prof. Chaomin Shen

09/2020-06/2023

- Led a project on accelerating the reconstruction of high-quality MR images from under-sampled k-space data. The project received the **3rd Prize** in the MRI reconstruction challenge at the 4th International Symposium on Image Computing and Digital Medicine.
- Led a project on key points detection in large volume CT scans, introducing a novel machine learning algorithm that has been deployed on a **surgical robot** to assist the preoperative analysis and diagnosis for hip replacement surgery.
- Presented a novel method Hybrid Atlas Building with Deep Registration Priors for efficient atlas building, which dramatically reduces time consumption. Related publications include [10].

Skills

Programming: Python, C/C++, JavaScript, CSS, and HTML, and have experience in MATLAB, CMake, Git, Scripting (Bash), LaTeX

Software: Pytorch , Linux, 3d Slicer

Selected Honors

2023	Academic Services awarded by UVA	VA USA
2022	Scholarship of STEM Cloud Center awarded by ECNU	Shanghai China
2022	Outstanding Student awarded by ECNU	Shanghai China
2022	“Fen Zhong” Scholarship awarded by ECNU	Shanghai China
2020	Academic Scholarship awarded by Ministry of Education	Shanghai China

Other Experience

Teaching Assistant	11/2021
The VIRTUAL and FREE Winter School in “AI in Image Registration” offered by Prof. Miaomiao Zhang (UVA)	
Invited Presentation	12/2021
Youth Seminar on Mathematical Theory and Methods in Image Processing	
Seminar Organizer	09/2020-07/2023
A weekly seminar for Medical Imaging Research Group at ECNU	
Teaching Assistant	Spring 2021
Mathematical Analysis, ECNU	
Web Developer, Sinosafe General Insurance Co. Ltd, Shenzhen, China	07/2017-06/2019
Developed and maintained web pages and mobile applications on a daily basis using JavaScript, HTML, CSS, and modern frameworks such as Vue.js and AngularJS. Those online platforms enable users to purchase insurance, manage orders, renew existing policies, and access a variety of related digital services.	